

Intention to use technology for micro-credential programs: evidence from technology acceptance and self-determination model

Miao Miao

School of Economics and Management, Chengdu Normal University, Chengdu, China

Mansoor Ahmed

Business School, Suzhou University, Suzhou, China

Noman Ahsan

Sindh Higher Education Commission, Karachi, Pakistan, and

Bushra Qamar

IQRA University, Karachi, Pakistan

Abstract

Purpose – The study aims to investigate students' attitudes toward using technology for micro-credential programs (MCPs) and their behavioral intention toward using MCPs for learning and enhancing their skills. The study also intends to investigate the moderating influence of labor market conditions (LMC).

Design/methodology/approach – A closed-ended questionnaire is employed to collect data from 474 participants through a convenience sample, from the university students in Karachi. Two theoretical frameworks are used in the study: technology acceptance model (TAM) and self-determination theory (SDT). The partial least squares structural equation modeling (PLS-SEM) technique is used to analyze data.

Findings – Findings reveal significant and positive relationships between all variables, except controlled motivation (CM) and perceived challenges. Moreover, the moderation results ascertain that the labor market does not moderate the relationship between attitude toward using technology for MCPs and students' behavioral intention toward using MCPs for learning.

Originality/value – The study seeks to understand students' attitudes and behavioral intentions toward using technology for MCPs. Also, the moderating effect of LMC is highlighted in understanding the impact of the attitude to use technology (AT) for MCPs and behavioral intentions in higher educational institutions (HEIs) in Pakistan. The study highlights intuitive practical implications for students of HEIs, universities and digital credential program providers.

Keywords Attitude to use technology, Micro-credential program, Higher education institution, Behavioral intention, TAM, Self-determination

Paper type Research paper

1. Introduction

Over the last decade, technological advancement along with rapid utilization of laptops, cell phones, tablets, high-speed Internet and improved access has significantly created opportunities to increase the effectiveness of higher education. Thus, online education platforms have escalated significantly (Ouyang *et al.*, 2020). Online learning is no longer a

This paper forms part of a special section "Micro-credential as a Digital Enabler for Higher Education Ecosystems", guest edited by Muhammad Ali, Syed Ali Raza, Chin Hong Pua and Wasim Qazi.

Funding: This research was funded by the Fujian Province Natural Science Foundation Project (No: 2022J01380), Fujian Province Social Science Foundation (No: FJ2022B088) and Sichuan Liquor Development research center (No: CJY21-07).



unique phenomenon in our educational system (Bowers and Kumar, 2015). Subsequently, massive open online programs have changed students' learning styles in post-secondary education through online video lectures, access to curricula, discussion forums and subject experts (Belawati, 2014; Dehghani *et al.*, 2020). Students prefer convenience in terms of reading, completing assignments, watching study-related videos and listening to lectures due to their hectic schedules. Higher education institutions (HEIs) are offering micro-credential programs (MCPs) for students to gain knowledge, improve their abilities and expertise (Gauthier, 2020). Some institutions have evolved towards online degree programs and credentialing forums. They also offer some granting academic credit and micro-credentials badges (Clinefelter and Aslanian, 2016; Shah *et al.*, 2019). Several experts have regarded this as a culmination of an advanced era of higher education.

According to Orman *et al.* (2023) digital technology breakthroughs and their effects in numerous disciplines present innovations and challenges to educational organizations. Digital technologies have become more crucial and even necessary in keeping the expertise and abilities of individuals up to date and expanding their capacity for lifelong learning resulting in increased use of modern technologies in our daily lives, especially in education. Many courses are now organized to increase the possibilities of lifelong learning. Micro-credentials largely shape the framework and validity of these courses and genuine certifications play an important role in the industry (Orman *et al.*, 2023). Employers and industries hope micro-credentials will provide a solution to labor market's rapid shift in requirements, allowing lifelong learners to obtain 21st-century industry skills (Oliver, 2019; Qazi *et al.*, 2023).

Micro-credentials change society's perception of learning and alter educational institutions' obligations by contributing to the digital transformation of education (Wheelahan and Moodie, 2021). By taking part in brief, affordable and flexible online training courses, learners were able to advance their knowledge and abilities (Oliver, 2019). Micro-credentials take the shape of a diploma or certificate in a scaled-down modular format. According to Milligan and Kennedy (2017) a micro-credential is a skill or level of proficiency demonstrated by a certificate, nano-degree, micro-master, badge, or membership. With micro-credentials, learning is shaped more rapidly with a flexible structure that is geared toward the demands of more people. The widespread change in learning is facilitated by learners' simple access to digital learning resources like online tutorials, digital books, video courses and other learning websites (Orman *et al.*, 2023). The transition from classroom learning to lifetime learning brings about a new system change in education and freedom from location and time constraints. This modification attempts to keep students abreast of societal and technical advancements (Schwab, 2016).

According to Pickard *et al.* (2018), micro-credentials are better than other online learning platforms as they offer various technical courses on diverse themes with certificates and credentials. Furthermore, micro-credentials offer a wide range of courses in different disciplines. Micro-credentials have acquired more attention after the flare-up of coronavirus disease 2019 (COVID-19). Globally, researchers have explored the influence of technology on learning, teaching and educational development by using diverse methods, techniques, factors and ideas (Meet and Kala, 2021). Moreover, higher education institutions, students, professors and educational institutions take advantage of MCPs and standardize their teaching and learning methodologies.

Micro-credentials have evolved as an important component of learning and professional growth that allows students to gain specialized knowledge and skill effortlessly (Wheelahan and Moodie, 2021). Micro-credentialing has been acknowledged by providers, firms and governments as a response to the volatility and velocity of change in labor markets, workplace capabilities and the demands of the 21st-century lifelong learner (Healy, 2021). Micro-credentials have gained popularity due to their potential to enhance employability and career advancement; as they are focused on specific technical areas (Brown *et al.*, 2021). Concurrently, a growing interest is observed in the utilization of technology in MCPs to provide learners with easy access and interactive learning experiences.

Despite the increasing integration of technology in education, [Kiiskila et al. \(2022\)](#) validate a significant gap in the students' intention to use technology for MCPs. Although several research studies have analyzed the acceptability and technology integration in education, specific features (such as self-guide, digital badges, flexibility and certifications) and motives (such as highly accessible, economical learning opportunities and e-learning) of micro-credentials have not been investigated. Thus, this study will investigate students' intention to use technology in MCPs. Additionally, [Haleem et al. \(2022\)](#) ascertain that examining students' motivation and desire to use technology in the classroom is crucial to evaluate the effectiveness of the program. Therefore, the current study will be an initial attempt to develop more efficient and learner-centric technology-enabled MCPs by examining factors that affect students' intentions to use technology in MCPs. Furthermore, [Oliver \(2019\)](#) also emphasized insufficient research conducted on micro-credentials and their variants. The study also confers Micro-credentials as a relatively new notion in education and learning.

According to [Prensky \(2009\)](#) the current generation of students is technologically dependent, as they are already tech-savvy ([Patil and Undale, 2023](#)). However, [Davis \(1989\)](#) claims students perceive technology with perceived ease of use (PEOU) rather than perceived usefulness (PU). Despite being technologically dependent, students believe technology is easy to use, but they might ignore how useful it is in the context of MCPs. This disparity is significant because if students do not perceive the potential benefits and value of technology in micro-credentials, they may not fully participate or maximize their learning experience ([Marshall and Marshall, 2018](#)). As a result, students will not focus on the benefits that come with it, such as increased creativity, professional growth, information acquisition, skill and expertise enhancement and lifetime learning ([Xu et al., 2022](#)). Hence, it is necessary to carry out this study to bridge the knowledge gap and acquire important information about students' attitudes to using technology for MCPs. Students' attitudes can identify potential constraints or challenges they may face when using technology for micro-credentials ([Thi Ngoc Ha et al., 2022](#)). Addressing these constraints or challenges, whether they are linked to accessibility, usability, or personal beliefs, may contribute to more successful technology integration while also encouraging lifelong learning and professional progress.

The current research adds to the academic literature in the following ways. Firstly, this study presents a comprehensive analysis of students' attitudes toward using technology for MCPs and their behavioral intention toward using MCPs for learning and enhancing their skills. Secondly, this study has targeted the students of private sector universities because private sector universities focus on MCPs, so it was easy to collect data from these universities. Also, private sector universities usually have stronger ties to companies and employers. These universities may strengthen their contacts with relevant firms and increase students' networking opportunities by giving micro-credentials targeted to specific industry demands ([Miller and Jorre De St Jorre, 2022](#)). Thirdly, this study will benefit academic institutions by providing comprehensive insight into students' attitudes and behaviors toward using technology for MCPs. Additionally, it also examines the moderating role of labor market conditions (LMC) on the relationship between attitudes to use technology for MCPs and behavioral intention in higher educational institutions (HEIs) in Pakistan. However, a major contribution of this study is the adoption of two models, i.e. the technology acceptance model (TAM) ([Davis, 1989](#)) and the Self-determination model ([Ryan and Deci, 2000](#)). Previous studies related to MCPs have observed lack of simultaneous focus on these two theories to explore student's attitudes and intentions to use technology for MCPs.

Many questions have been raised about micro-credentials, their significance, quality authentication and inclusion in formal educational contexts and how to incorporate them into traditional education systems, particularly those in higher education institutions. The research questions of this study are (1) Do students' attitudes toward using technology for MCPs enhance their behavioral intention? (2) Do LMC moderate the role of students' attitudes towards using technology for MCPs to enhance their behavioral intention?

The following is how paper is structured: [Section 2](#) offers review of theoretical framework and hypotheses literature. [Section 3](#) discusses methodology and data collecting, while [Section 4](#) discusses study findings and discussion. Finally, [Section 5](#) includes a conclusion, limitations and future recommendations.

2. Literature review

2.1 Theoretical background

To explain the diverse structural links developed in this study, the current research relied on two theories: TAM and self-determination theory (SDT).

TAM model arose from [Fishbein and Ajzen's \(1977\)](#) theory of reasoned action (TRA), which was further expanded by [Davis \(1989\)](#), which is by far the most widely used framework involving attitudes and technology adoption. This theoretical framework has been utilized in every research to investigate factors that influence how people use and accept new technology ([Venkatesh and Davis, 2000](#); [Al Kurdi et al., 2020](#)). [Davis \(1989\)](#) states the relationship between opinion, perspective, belief, behavior and purpose, in the technology adoption model allows us to predict how novel technologies will be used. The TAM model may be viewed as a crucial component in the execution of the information system (IS) and understanding of user acceptability ([Bailey et al., 2022](#)). The model successfully predicts how agreeable a tool will be and acknowledges the additional changes that must be made to the framework for users to find it acceptable ([McNamara and Sepasgozar, 2020](#)). According to [Fernandes and Oliveira \(2021\)](#) TAM is a widely accepted theory that largely focuses on how a person mentally interacts with new developing technologies. TAM helps us better understand how people utilize technology ([Rad et al., 2022](#)). The TAM model, according to [Nandi and Mehendale \(2002\)](#) depicts factors such as PEOU, PUn, attitude, actual use and behavioral intention. Researchers believe that PEOU and PUn are the two features most likely to impact users' attitudes toward using virtual-based learning ([Davis et al., 1989](#); [Nandi and Mehendale, 2022](#)). PEOU is the degree to which a person believes a system to be simple to use ([Raza and Khan, 2022](#)). PUn, is the degree to which a person believes that applying a system would improve performance ([Raza and Khan, 2022](#)). The TAM model has been extensively used in several studies ([Venkatesh and Davis, 2000](#); [Thong et al., 2006](#); [Venkatesh et al., 2003](#); [Lee et al., 2007](#)). TAM doesn't take into account demographic, external and economic factors that have limited applicability of this model in assessing a person's attitude and intention toward adopting technology for MCPs. According to [Davis \(1989\)](#), TAM permits inclusion of external variables such as PU and PEOU determinants. Additionally, any significant attempt to increase technology employment can succeed or fail based on how quickly individuals adopt new technology ([Sharma et al., 2017](#)). The TAM model is still suitable for analyzing students' attitudes toward technology integration in the educational system for learning and upgrading their skills ([Davis et al., 1989](#)). Hence, this study provides a better understanding of students' attitudes toward using technology for MCPs and increases collaborative learning, which will improve students' behavioral intentions ([Al-Rahmi et al., 2014](#)).

Several theories have been proposed on the idea of motivation ([Middleton and Toluk, 1999](#); [Marshall and Marshall, 2018](#)) but one of the best is [Deci and Ryan \(1985\)](#) SDT. According to this theory, self-motivation is a method that focuses on the development of personality and behavior regulation ([Ryan et al., 1997](#); [Ryan and Deci, 2000](#)). The SDT theory has attracted the interest of numerous researchers who have implemented it in a variety of research scenarios since it is thought to be a valuable theory for understanding an individual's motivation levels ([Deci et al., 1991](#); [Gagne and Deci, 2005](#)).

This theory describes motivation as a multidimensional variable and distinguishes between two types of motivation: intrinsic (autonomous) motivation and extrinsic (controlled)

motivation (Deci and Ryan, 1985). The theory states that the type of behavior varies along the self-determination continuum and is dependent on the underlying behavior (Ryan and Deci, 2000). A person who has an honest desire acts completely freely and with no restrictions, which is the ultimate level of self-determination (Arvanitis and Kalliris, 2020). Extrinsic motivation, on the other hand, exhibits a limited level of self-determination and requires an individual to act within external restrictions to achieve particular results (Van den Broeck *et al.*, 2021). Several studies validate SDT influences individual behavior in various ISs such as massive open online courses (MOOCs) (Zhou, 2016), education (De Naeghel *et al.*, 2016), mobile-based assessment (Nikou and Economides, 2017) and online learning (Hartnett, 2015).

According to Raza *et al.* (2021), students who are intrinsically motivated will go to school to learn new information or skills. Extrinsically motivated students are those that behave for a reason, such as going to school because it's necessary for the job they want. Thus, SDT is employed in this study to assess whether students are intrinsically or extrinsically motivated toward using technology for MCPs and which would increase their collaborative learning thus improving students' behavior intentions.

Together, SDT and TAM have been demonstrated to be the foundations of user acceptance of new and emerging technologies across a variety of disciplines, including the intention to use e-learning continuously in the workplace, as stated by Roca and Gagné (2008) and acceptance of learning via a mobile device, as noted by Nikou and Economides (2017). Both TAM and SDT take alternative approaches to explain students' attitudes and behaviors regarding their perception of using new and advanced technology (Racero *et al.*, 2020). Although each model has several advantages and disadvantages, they have different focal objectives; nonetheless, they cannot completely explain the behavioral intention of students. Simultaneously, in this study, TAM and SDT models examine the student's intention to use technology for MCPs. TAM emphasizes the cognitive aspects of students' behavior intentions while the SDT model focuses on functional features of new technology.

2.2 Hypothesis development

2.2.1 Perceived ease of use (PEOU). PEOU is a significant element influencing students' behavioral intention and attitudes toward technology integration for learning through different programs (Mailizar *et al.*, 2021). According to Davis (1989), PEOU is the degree to which a person believes adopting a given system will be effortless. Alternatively, it refers to the system's accessibility to information, adaptability, usability and ease of use. As a result, learners are more likely to utilize technology if they perceive it to be user-friendly (Al-Mamary, 2022; Ali *et al.*, 2022b). Thus, students are more inclined to embrace technology for learning when it is effortless and easy to use (Al-Marroof *et al.*, 2020). Prior research has also validated that PEOU positively influences attitude to use technology (AT) (Wong *et al.*, 2012); particularly in online learning environments (Teo *et al.*, 2018). Moreover, students are more likely to demonstrate positive attitudes about using technology if it is easy to use (Songkram *et al.*, 2023). Studies have established a strong correlation between attitudes toward using technology and PEOU. Numerous studies have revealed that PEOU significantly affects attitude (Davis *et al.*, 1989; Lust *et al.*, 2012; Abdullah and Ward, 2016). Therefore, the new hypotheses are put forth:

H1. PEOU significantly influences attitudes toward using technology for MCPs.

2.2.2 Perceived usefulness (PUn). The term PUn refers to the degree to which students think e-learning will enhance academic results (Lee, 2009). PUn is a significant aspect influencing students' behavioral intentions, as PUn increases their learning performance, collaboration and efficiency with instructors and coworkers (Virani *et al.*, 2023). Al-Okaily *et al.* (2020) affirm that PUn influences students' intentions to use the e-learning system during certain

circumstances, i.e. COVID-19. Furthermore, [Alghizzawi et al. \(2019\)](#) discovered that PUn and PEOU are primary processes promoting e-learning adaptation and utilization. Prior research has found a positive association between PUn and students' intention to use a system ([Sabah, 2020](#)). In contrast, PUn also positively influences students' attitudes toward using technology for MCPs ([Abdullah and Ward, 2016](#); [Ifinedo, 2018](#)). According to [Chen and Tseng \(2012\)](#) PUn influences students' attitudes and intentions toward using technology for learning. Thus, students who perceive technology as a useful component demonstrate a more positive intention to use it for MCPs ([Kiiskilä et al., 2023](#)). As a result, we propose the following:

H2. PUn significantly influences attitudes toward using technology for MCPs.

2.2.3 Autonomous motivation (AM). The key concept of SDT is autonomous motivation (AM), defined as one's behavior manifests a perceived cause or purpose ([Ryan and Connell, 1989](#)). [Wang and Hou \(2015\)](#) claim AM is more effective at initiating certain behaviors because it is associated with successful outcomes while controlled motivation (CM) is related to negative consequences. According to [Rodrigues et al. \(2023\)](#) self-motivated individuals are more consistent with their behavior in the absence of extrinsic benefits, which produces adaptable results (i.e. satisfaction and competence). AM has been investigated in a variety of aspects, such as e-learning continuance intention, behavioral intention ([Al-Marroof et al., 2020](#)), student engagement ([Lai et al., 2021](#)) and discontinuous intention ([Luqman et al., 2018](#)). In addition, research has identified personal motivation as a significant factor in the attitude to utilize technology for learning, entertainment, or upgrading skills ([Reinecke et al., 2014](#)), etc. Previous studies have concluded that AM positively affects the AT ([Hagger and Chatzisarantis, 2016](#); [Zhou, 2016](#)). Thus, the following hypothesis is formulated:

H3. AM significantly influences attitudes toward using technology for MCPs.

2.2.4 Controlled motivation (CM). [Ryan and Deci \(2000\)](#) categorized internal and external motivation as two types of CM. Internal motivation depicts persons' behavior when they are engaged in activities for self-determining motives such as avoidance of guilt, obligation, internal rewards and ego-enhancement ([Nakayoshi et al., 2021](#)). In contrast, external motivation illustrates how people behave when they are engaged in a certain activity for reasons established by other mechanisms, such as external rewards, compliance and avoiding punishments ([Gammma et al., 2021](#)). According to SDT, external motives are the most tightly regulated extrinsic incentive, while internal motives are more self-determined ([Ryan and Deci, 2000](#)). People who can control motivation will be consistent in their behavior due to external reinforcement, pressure, or obligation ([Wu and Li, 2019](#)). Thus, CM has certain similarities with developing individual motivation and its influence on individual intention towards using technology ([Chiu and Cho, 2021](#)). CM suggests that students complete tasks under stress as they may experience mixed emotions and be less hesitant to adopt new technology ([Grant et al., 2011](#)). Thus, students are enthusiastic to perform the tasks that are considered mandatory ([Ryan and Deci, 2020](#)). Accordingly, students with CM are less likely to use technology for learning ([Alamer and Al-Khateeb, 2023](#)). Previous research revealed CM negatively impacts an individual's ATU ([Luqman et al., 2018](#)). Thus, the following hypothesis is formulated:

H4. CM significantly influences attitudes toward using technology for MCPs.

2.2.5 Perceived challenges. Numerous research studies have addressed the issues connected with technology-based learning ([Cheng and Lai, 2020](#); [Aboagye et al., 2021](#)). Technology-based learning is challenging because it requires students to manage time, organize and monitor their progress, upgrade their skills and regulate their learning ([Rapanta et al., 2020](#)). Students' attitudes toward using innovative technology incite enjoyment, motivation, learning and anxiety ([Aguilera-Hermida, 2020](#)). These psychological characteristics can influence students' attitudes toward using technology for online learning ([Wang et al., 2021](#);

Yu, 2021a, b). According to [Aboagye et al. \(2021\)](#) the student perspective toward using technology in various programs for learning has proven detrimental because of high dropout rates and lack of motivation. Other challenges included low student satisfaction, communication gaps, lack of desire and a dearth of real-time feedback ([Looi et al., 2021](#)). Similarly, [Almaiah et al. \(2020\)](#) recognized critical aspects and challenges of using technology in e-learning systems. [Alea et al. \(2020\)](#) studied students' perceptions regarding adopting technology for learning; findings discovered numerous problems and specific issues related to online learning. Additionally, [Abbasi et al. \(2020\)](#) observed that learners favor physical learning more than using technology in the classroom. Finally, [Favale et al. \(2020\)](#) examine students' emotions and sentiments about learning via technology in various programs. The following hypotheses are formulated from the above discussion:

H5. Perceived Challenges significantly influence attitudes toward using technology for MCPs.

2.2.6 Perceived learning (PL). Perceived learning (PL) signifies an improvement in students' perception regarding the level of knowledge and aptitude before and after the learning experience ([Alavi et al., 2002](#); [Attard and Holmes, 2022](#)). PL analyzes students' accomplishments and achievements after the completion of any course or program ([Kennedy, 2006](#)). Several researchers proposed that individuals' attitude toward technology directly affects academic performance in an online learning program ([Lin et al., 2019](#); [Aguilera-Hermida, 2020](#)). Similarly, [Perez-Sanagustn et al. \(2016\)](#) observed students' academic performance was improved during engagement in course-related activities posted in discussion forums. [Sebastianelli et al. \(2015\)](#) state well-structured and well-organized content enhances students' PL. Students' perceptions about learning are augmented through new programs by active participation in online learning ([Vayre and Vonthron, 2019](#)). [Lan and Hew \(2020\)](#) state learning engagement (in terms of emotions, cognition and behavior) can forecast the level of student learning in Micro-credentialing programs. [Panigrahi et al. \(2021\)](#) explored effect of PL on students' attitudes regarding the use of technology in online learning environments in the context of Indian higher education. However, minimal literature is available regarding students' perceptions of learning and attitudes toward using technology in MCPs ([Navanitha et al., 2022a, b](#)). Hence, Therefore, we proposed following hypotheses:

H6. PL significantly influences attitudes toward using technology for MCPs.

2.2.7 Attitude to use technology (AT). An individual's perspective and understanding (both good and negative) of their capacity to engage in particular conduct is referred to as their attitude ([Fishbein and Ajzen, 1977](#); [Chatterjee et al., 2021](#)). However, their attitude to adopt new technology is determined by behavior ([Mital et al., 2015](#)). [Hunt \(1977\)](#) established that the AT is an emotional aspect created from cognitive ideas (e.g. PU and PEOU); primarily, it is a composite function of past attitudes and intentions created from their experience with the system. In addition, attitude can be used to assess someone's intention to adopt technology and determine their action ([Sulaiman et al., 2012](#); [Chatterjee et al., 2021](#)). AT greatly affects students' behavioral intentions ([Abdullah and Ward, 2016](#)). Previous studies validate that attitude can easily measure users' behavioral intention to utilize new technology ([Davis, 1989](#); [Taylor and Todd, 1995](#); [Chatterjee et al., 2021](#)). Thus, the relationships between attitudes and behavioral intention have received considerable attention in behavioral research ([Chen et al., 2021](#)). Existing research considers attitude as a strong predictor of intention to do an action and represents an individual's judgment over the behavior under consideration ([Sahu et al., 2020](#)). However, attitudes can be either positive/negative depending on how an individual perceives using technology for online learning programs such as micro-credentials ([Hartwig et al., 2022](#)). A positive or favorable attitude is considered complementary to actual behavior

realization via the intention to undertake the behavior (Petty *et al.*, 2020). Many research studies have highlighted association among student attitude and e-learning behavior (Davis *et al.*, 1989; Queiroz and Wamba, 2019). Scherer *et al.* (2019), Tamilmani *et al.* (2021) reviewed association among attitudes and behavioral intentions; the findings suggest students' intentions regarding their behaviors on an idea depict their attitude. Recent research confirms the favorable relationship between attitudes and behavioral intention in adopting technology for learning (Al Lily *et al.*, 2020; Yu, 2021a, b; Camilleri, 2021). The following hypothesis is developed:

H7. Attitude towards using technology significantly influences Behavioral Intention

2.2.8 Labor market conditions (LMC). The labor market, also known as the employment market, is a place where employers and employees tradeoff between providing and requesting labor (Shao, 2022). It is very important in any economy, as it is closely related to products, markets for capital and services (Kenton, 2023). LMC refer to the situation of the labor market, which includes components such as the rate of employment, availability of jobs and demand for particular skills (Bejaković and Mrnjavac, 2020). Various studies have used LMC in their research as they influence individuals' decisions and behaviors regarding education, training and skill development (Qenani *et al.*, 2014; Sima *et al.*, 2020). Fu *et al.* (2021) in their study indicated that positive LMC, such as low unemployment and high job demand in some industries, can provide economic incentives for individuals to invest in additional education and skill development. Moreover, students perceive LMC as the initial step in determining their long-term career opportunities. According to Navanitha *et al.* (2022a, b), students demonstrate high behavioral intention and actively participate in technology-based MCPs during favorable market conditions, characterized by high employment demand and multiple opportunities. Students believe that participation in MCPs will assist them in developing the relevant and demanding skills and abilities mandatory for future job opportunities and prospects. Extensive research exhibit that graduate students who enter the workforce during poor macroeconomic conditions suffer considerable losses (Arellano-Bover, 2022; Haines *et al.*, 2010 state that students with refined skills and knowledge can search for alternative employment opportunities in the labor market than students with lower knowledge and capabilities. Littlejohn and Hood (2019) investigated the use of technology for micro-credentialing programs and their impact on behavioral intention and conclude that behavioral intentions may vary depending on the condition of the labor market. Voßemer *et al.* (2018) As labor market insecurities have been growing it has been observed that unemployment and insecure jobs negatively affect individuals' health and well-being. Therefore this study has adopted the moderating role of LMC to investigate students' attitudes toward using technology for MCPs and their behavioral intention toward using MCPs for learning and enhancing their skills. Furthermore, students who are familiar with technology utilization have greater opportunities and try to enhance it by making it a behavioral action. Thus, we propose the following hypothesis:

H8. LMC moderate the relationship between attitude towards using technology for MCPs and behavioral intention

3. Research methodology

3.1 Research model

Figure 1 demonstrates the conceptual framework of the present study. The model explains relations between independent and dependent variables. The framework includes factors (PUn, PEOU, CM, AM, PL and perceived challenges) that affect students' attitudes toward

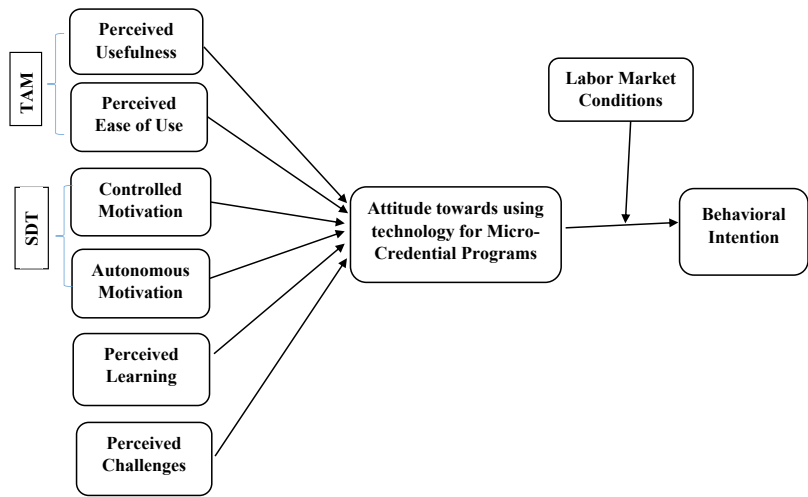


Figure 1.
Conceptual framework

Source(s): Authors' own work

technology-based MCPs by enhancing their behavioral intention. In addition, the study also highlights moderating effects of LMC that could enhance students' attitudes and intention to use technology for MCPs.

3.2 Data collection and research instrument

This study utilizes a quantitative approach to investigate students' attitudes toward using technology for MCPs and their behavioral intention to utilize such programs for learning and enhancing their skills with a moderating effect of LMC. Quantitative research is typically connected with a logical method that tests theories using evidence (Morgan, 2022). Data was collected from the students of private sector universities by using a convenience sample approach. The convenience sampling method is a non-probability strategy to collect data from people who are easily accessible (Qureshi *et al.*, 2023). We evaluated private institutions to attain an understanding of students' attitudes and behaviors across different demographics because private universities have an extensive student body, including individuals from various socioeconomic origins, ethnicities and academic fields. Furthermore, private educational institutions offer specialized programs that may correspond to certain MCPs. Therefore, researchers can analyze the suitability and usefulness of these technology-based programs. A total of 500 questionnaires were mailed to students via social media and after removing the observations with missing values, 474 questionnaires were considered valid and suitable for analysis. As a result, this study's sample size is 474 students. A self-administered questionnaire with closed-ended questions was adapted to collect data from the respondents. A five-point Likert scale ranging from strongly disagree (1) to strongly agree (5) was used to collect data. A survey questionnaire was adapted from several published studies (see Table 1). We conduct pilot testing and the results (of a sample-like population) suggest that our instrument is easy to understand by the respondents.

3.3 Demographics

Table 2 represents the respondent profiles which are described as per the demographic analysis. Firstly, the gender distribution of the responses revealed that males were the

				Technology for micro-credential programs
S.NO	Variable	Type	Source	
1	Behavioral intention	Dependent variable	Pappas <i>et al.</i> (2017)	
2	Labor market conditions	Moderator	Alvarez-Gonzalez <i>et al.</i> (2017)	
3	Perceived use	Independent variable	Davis (1989), Venkatesh and Bala (2008)	
4	Perceived ease of use	Independent variable	Davis (1989), Zhang and Benyoucef (2016)	
5	Controlled motivation	Independent variable	Zhou (2016)	
6	Autonomous motivation	Independent variable	Zhou (2016)	
7	Perceived learning	Independent variable	Peltier <i>et al.</i> (2003)	
8	Perceived challenges	Independent variable	Musingafi <i>et al.</i> (2015)	
9	Attitude toward using technology for micro-credential program	Independent variable	Taylor and Todd (1995)	

957

Table 1.
Variables description

Demographic items	Frequency	Percentile
<i>Gender</i>		
Male	260	54.9%
Female	214	45.1%
<i>Age</i>		
20 or Less than 20	33	7.0%
21–25	127	26.8%
26–30	177	37.3%
36–40	97	20.5%
41 or above	40	8.4%
<i>Education</i>		
Undergraduate	149	31.4%
Graduate	199	42.0%
Postgraduates	111	23.4%
Other	15	3.2%

Source(s): Authors' own work

Table 2.
Respondent's
profile (N = 474)

majority of the participants totaling up to 54.9% while the percentage of women was 45.1. In terms of respondents' ages, 7.9% of the population was under the age of 20 and less, 26.8% were between the ages of 21 and 24, 37.3% were between the ages of 26 and 30, 20.5% were between the ages of 36 and 40, while only 8.4% were over the age of 41 and above. In addition, the respondents' educational backgrounds revealed that 42% of the respondents were graduates the highest percentage, followed by undergraduates (31.4%), postgraduates (23.4%) and others (3.2%).

4. Data analysis and results

Smart-PLS, partial least squares structural equation modeling (PLS-SEM) technique was used in current investigation (Ringle *et al.*, 2015; Ali and Raza, 2017). Therefore, we used 5,000

subsamples of the bootstrapping approach to calculate significance level for each path coefficient. Two stages were taken to complete PLS-SEM. The evaluation of the measurement model occurs first and evaluation of the structural model occurs second. We evaluated construct validity and discriminant validity criteria for the measurement model and we also evaluated the importance of path coefficients for the structural model. According to [Ringle et al. \(2015\)](#), SEM is a good statistical technique; as it examines the validity of the study's hypothesis by statistical data and figures. The hypothetical model is examined using a variance-based methodology in the current research. An appropriate approach to study and examine integrated models in diverse contexts for research is to implement PLS-SEM utilizing Smart-PLS software ([Chin, 1998](#); [Ali et al., 2022a](#)).

4.1 Measurement model

The measurement model examines the connection between latent constructs and measures. Various tests were performed in this study to investigate validity and reliability, as suggested by [Hair et al. \(2011\)](#). There are two aspects to the measurement model: convergent validity and discriminant validity. The average variance extracted (AVE), Cronbach's alpha, loadings and composite reliability, are used to calculate convergent validity. Furthermore, Cronbach's alpha and the composite reliability scale are used to assess reliability. In contrast, the heterotrait-monotrait (HTMT) correlation ratio is employed to evaluate discriminant validity.

Three measures for evaluating convergent validity are Cronbach's alpha, composite reliability and AVE. Cronbach alpha evaluates data dependability as well as survey item internal consistency. Internal consistency reliability is the amount to which all items on a certain subscale measure the same notion ([McCrae et al., 2011](#)). [Table 3](#) shows that the value of Cronbach's alpha for all variables is larger than 0.7, which meets Nunnally and Bernstein's criterion (1994). Furthermore, the reliability of items may be tested using composite reliability and AVE. This can be seen in the table; all variables have simultaneously achieved an appropriate composite reliability value (i.e. more than 0.7) as proposed by [Fornell and Larcker \(1981\)](#). Furthermore, [Hair et al. \(2014\)](#) demonstrate that all AVE values of variables are greater than 0.5, indicating criterion is met and convergent validity is achieved ([Raza et al., 2019](#)).

As convergent validity is examined, the model's discriminant validity is explored, which refers to the degree to which a latent variable differentiates from other variables ([Carmines and Zeller, 1979](#)) and is examined by a test suggested by [Fornell and Larcker \(1981\)](#). As indicated in [Table 4](#), all diagonal values of the discriminant validity matrix (which represent the square root of AVE and should be bold) are greater than off-diagonal values, indicating that discriminant validity has been established. Furthermore, [Table 5](#) represents the cross-loadings of items. The findings highlight that each item is more highly loaded in its relevant construct than related variables. Additionally, the cross-loading difference exceeds the 0.1 recommended level ([Ali et al., 2017](#); [Gefen and Straub, 2005](#)). Moreover, we evaluated the HTMT ratio to discover the scattered association between the indicators. As seen in [Table 6](#), all table values are not more than 0.85 or 0.9, as suggested by [Gold et al. \(2001\)](#); hence discriminant validity is achieved. Finally, measurement model analysis has demonstrated the scale's reliability and validity. Therefore, proceeding to the structural model is reliable and valid because the framework's distinctiveness has been confirmed.

4.2 Structural model

The structural model analysis was performed using standardized routes to obtain the results of the proposed hypothesis. [Table 7](#) displays findings of first-order analysis while [Table 8](#) displays findings of the moderating variable that was tested. Moreover, [Figure 2](#) depicts the results of standardized regression weight (SRW). *P*-value should not be more significant than 0.01 (1%), 0.05 (5%) or 0.1 (10%) for a hypothesis to be accepted. As demonstrated in [Table 7](#),

	Items	Loadings	Cronbach's alpha	Composite reliability	Average variance extracted
AM	AM1	0.769	0.764	0.863	0.677
	AM2	0.847			
	AM3	0.850			
AT	AT1	0.794	0.869	0.910	0.718
	AT2	0.852			
	AT3	0.874			
	AT4	0.867			
BI	BI1	0.862	0.870	0.920	0.794
	BI2	0.913			
	BI3	0.896			
CM	CM1	0.835	0.838	0.902	0.755
	CM2	0.892			
	CM3	0.879			
LMC	LMC1	0.702	0.937	0.929	0.725
	LMC2	0.882			
	LMC3	0.890			
	LMC4	0.867			
	LMC5	0.899			
PC	PC1	0.909	0.852	0.910	0.771
	PC2	0.851			
	PC3	0.874			
PEOU	PEOU1	0.826	0.796	0.880	0.710
	PEOU2	0.837			
	PEOU3	0.864			
PL	PL1	0.780	0.775	0.856	0.597
	PL2	0.744			
	PL3	0.796			
	PL4	0.770			
PU _n	PU _n 1	0.839	0.873	0.913	0.724
	PU _n 2	0.854			
	PU _n 3	0.882			
	PU _n 4	0.828			

Note(s): AM = autonomous motivation; AT = attitude to use technology; BI=behavioral intention; CM=controlled motivation; LMC=labor market conditions; PC=perceived challenges; PEOU=perceived ease of use; PL=perceived learning; PU=perceived usefulness

Source(s): Authors' own work

Table 3.
Measurement model
results

	AM	At	BI	CM	LMC	PC	PEOU	PL	PU
AM	0.823								
AT	0.472	0.847							
BI	0.370	0.701	0.891						
CM	0.370	0.538	0.479	0.869					
LMC	0.006	0.069	0.070	0.079	0.851				
PC	0.404	0.267	0.206	0.295	-0.069	0.878			
PEOU	0.373	0.516	0.470	0.633	0.066	0.291	0.843		
PL	0.630	0.552	0.487	0.497	-0.040	0.493	0.455	0.773	
PU _n	0.326	0.573	0.505	0.669	0.020	0.243	0.588	0.452	0.851

Note(s): AM = autonomous motivation; AT = attitude to use technology; BI=behavioral intention; CM=controlled motivation; LMC=labor market conditions; PC=perceived challenges; PEOU=perceived ease of use; PL=perceived learning; PU=perceived usefulness

Source(s): Authors' own work

Table 4.
Fornell–Larcker
criterion

	AM	AT	BI	CM	LMC	PC	PEOU	PL	PU
AM1	0.769	0.302	0.212	0.259	0.014	0.350	0.290	0.443	0.231
AM2	0.847	0.432	0.381	0.312	-0.016	0.344	0.313	0.560	0.295
AM3	0.850	0.412	0.297	0.334	0.021	0.315	0.317	0.539	0.272
AT1	0.348	0.794	0.578	0.441	0.091	0.215	0.408	0.454	0.443
AT2	0.433	0.852	0.610	0.431	0.070	0.202	0.425	0.479	0.418
AT3	0.409	0.874	0.597	0.462	0.044	0.248	0.440	0.471	0.540
AT4	0.407	0.867	0.591	0.490	0.031	0.238	0.473	0.465	0.537
BI1	0.282	0.631	0.862	0.465	0.034	0.160	0.465	0.392	0.488
BI2	0.373	0.636	0.913	0.411	0.074	0.208	0.393	0.464	0.432
BI3	0.332	0.604	0.896	0.401	0.079	0.183	0.398	0.444	0.431
CM1	0.280	0.443	0.431	0.835	0.065	0.228	0.516	0.427	0.621
CM2	0.363	0.518	0.421	0.892	0.049	0.283	0.604	0.475	0.547
CM3	0.314	0.436	0.395	0.879	0.096	0.254	0.523	0.387	0.584
LMC1	-0.009	0.004	-0.002	0.009	0.702	-0.053	0.020	-0.082	-0.053
LMC2	0.003	0.027	0.033	0.047	0.882	-0.075	0.047	-0.071	0.003
LMC3	0.016	0.027	0.031	0.040	0.890	-0.077	0.043	-0.077	0.000
LMC4	0.034	0.029	0.028	0.024	0.867	-0.061	0.036	-0.068	-0.029
LMC5	-0.006	0.094	0.092	0.102	0.899	-0.052	0.075	-0.001	0.042
PC1	0.377	0.238	0.164	0.241	-0.074	0.909	0.236	0.451	0.187
PC2	0.370	0.203	0.145	0.195	-0.012	0.851	0.198	0.380	0.089
PC3	0.324	0.256	0.227	0.327	-0.087	0.874	0.319	0.460	0.338
PEOU1	0.224	0.460	0.407	0.536	0.037	0.214	0.826	0.355	0.574
PEOU2	0.358	0.412	0.366	0.528	0.070	0.274	0.837	0.404	0.437
PEOU3	0.368	0.429	0.413	0.536	0.064	0.250	0.864	0.394	0.467
PL1	0.408	0.438	0.399	0.406	-0.060	0.532	0.357	0.780	0.351
PL2	0.416	0.387	0.371	0.300	-0.063	0.423	0.258	0.744	0.273
PL3	0.568	0.453	0.392	0.431	-0.003	0.286	0.391	0.796	0.392
PL4	0.549	0.424	0.342	0.388	-0.002	0.290	0.392	0.770	0.372
PUn1	0.239	0.474	0.413	0.572	-0.018	0.153	0.530	0.343	0.839
PUn2	0.319	0.478	0.463	0.586	0.059	0.248	0.489	0.384	0.854
PUn3	0.274	0.498	0.404	0.575	0.054	0.217	0.495	0.368	0.882
PUn4	0.277	0.498	0.440	0.545	-0.026	0.210	0.487	0.440	0.828

Table 5.
Loadings and cross-loadings

Note(s): AM = autonomous motivation; AT = attitude to use technology; BI=behavioral intention; CM=controlled motivation; LMC=labor market conditions; PC=perceived challenges; PEOU=perceived ease of use; PL=perceived learning; PU=perceived usefulness
Source(s): Authors' own work

	AM	At	BI	CM	LMC	PC	PEOU	PL	PU
AM									
AT	0.568								
BI	0.441	0.806							
CM	0.454	0.628	0.560						
LMC	0.033	0.048	0.050	0.062					
PC	0.507	0.307	0.236	0.341	0.078				
PEOU	0.481	0.618	0.564	0.771	0.061	0.349			
PL	0.807	0.671	0.592	0.609	0.083	0.605	0.577		
PUn	0.395	0.657	0.580	0.786	0.059	0.272	0.702	0.545	

Table 6.
Heterotrait-monotrait ratio (HTMT)

Note(s): AM = autonomous motivation; AT = attitude to use technology; BI=behavioral intention; CM=controlled motivation; LMC=labor market conditions; PC=perceived challenges; PEOU=perceived ease of use; PL=perceived learning; PU=perceived usefulness
Source(s): Authors' own work

Table 7.
Results of path
analysis

Hypothesis	Regression path	Effect type	β -Coeff	<i>P</i> values	Remarks
H1	PEOU $\rightarrow$ AT	Direct Effect	0.133	0.051	Supported
H2	PU _n $\rightarrow$ AT	Direct Effect	0.278	0.000	Supported
H3	AM $\rightarrow$ AT	Direct Effect	0.167	0.000	Supported
H4	CM $\rightarrow$ AT	Direct Effect	0.105	0.121	Not Supported
H5	PC $\rightarrow$ AT	Direct Effect	-0.054	0.163	Not Supported
H6	PL $\rightarrow$ AT	Direct Effect	0.234	0.000	Supported
H7	AT $\rightarrow$ BI	Direct Effect	0.695	0.000	Supported

Note(s): AM = autonomous motivation; AT = attitude to use technology; BI=behavioral intention; CM=controlled motivation; LMC=labor market conditions; PC=perceived challenges; PEOU=perceived ease of use; PL=perceived learning; PU=perceived usefulness

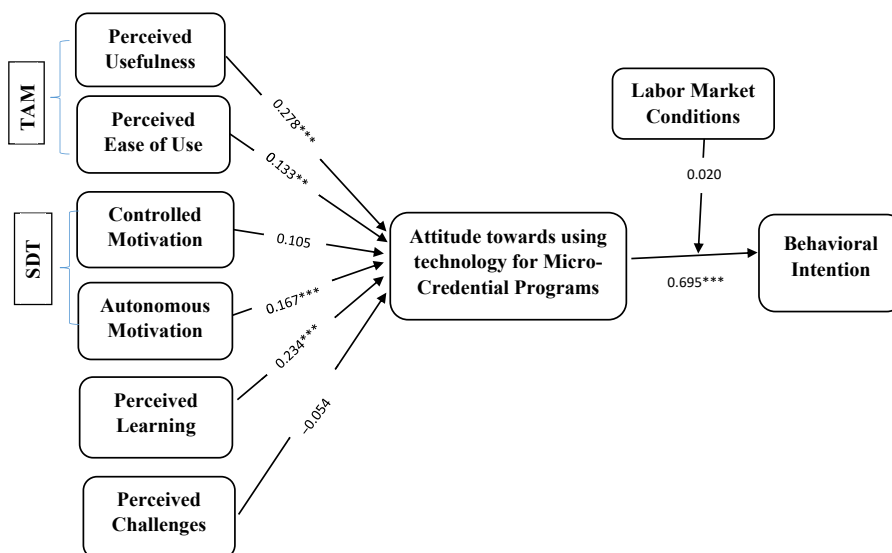
Source(s): Authors' own work

Table 8.
Moderating role of
labor market
conditions

Hypothesis	Regression path	Effect type	β -Coeff	<i>P</i> values	Remarks
H8	AT*LMC $\rightarrow$ BI	Indirect Effect	0.020	0.659	Not Supported

Note(s): AT = attitude to use technology; BI=behavioral intention; LMC=labor market conditions

Source(s): Authors' own work



Note(s): ** denotes significance at 5%, *** denotes significance at 1%

Source(s): Authors' own work

Figure 2.
Results of path
analysis

all determined hypotheses were accepted except for H3 and H4, where H3 shows a positive but insignificant impact, while H4 shows a negative and insignificant impact. In addition, the moderating effect of LMC demonstrates no moderation between the variables.

4.3 Discussion

Tables 7 and 8 exhibit results of first-order analyses. The findings support the first hypothesis, i.e. AM positively and significantly affects students' attitude toward using technology for MCPs as ($\beta = 0.167^{***}$; p -value < 0.01). The results align with a previous study by Zhou (2016) and Pozón-López *et al.* (2021). Their findings conclude students are more inclined towards technology and demonstrate a positive attitude towards using technology for MCPs. According to Ginting (2021) autonomously motivated students actively participate and engage in the learning process. They are therefore more likely to use technology to enrich their educational process and obtain micro-credentials. Autonomously motivated students perceive technology as a tool rather than a barrier to achieving their educational objectives. They are open to exploring and implementing technology with a positive attitude since they understand the advantages (continuous learning, skill development opportunities, accessibility and cost-effectiveness) of using it to obtain micro-credentials (Sabah, 2020).

The second hypothesis depicts a positive and significant relationship between attitude toward using technology and behavioral intention ($\beta = 0.695^{***}$; p -value < 0.01). The findings are consistent with the studies of Sadaf *et al.* (2012) and Huang *et al.* (2020). The findings reveal that students' behavioral intentions are highly influenced by their attitudes (likes and dislikes) toward embracing technologies for learning. As students' attitude toward technology gets more positive, their intention to use technology for MCPs increases (Navanitha *et al.*, 2022a, b). Students who regard technology as useful, productive and valuable are more likely to have a positive behavioral intention to utilize technology in various activities. Since, they perceive technology as a useful tool that can improve their performance, educational experience, or overall quality of life. This positive attitude determines individuals' motives, preferences and decisions toward the adoption of technology (Venkatesh *et al.*, 2012).

The third hypothesis shows a negative and insignificant relationship between CM and AT for MCPs ($\beta = 0.105$; p -value > 0.1). The findings corroborate the studies of Sabah (2020), Reed (2023). The results conclude that Students' intentions to use technology for MCPs are not influenced by CM. Coman *et al.* (2020) claim students who are forced or pressured to use technology for these programs may develop a more negative perspective of technology in their learning experiences. Hence, external factors including career opportunities, market demands and incentives influence students' motivation to use technology. It signifies that forcing students to use technology for micro-credentials will not yield more positive attitudes (Pollard and Vincent, 2022). However, intention to use technology for MCPs may be more successful if attempts are made to promote intrinsic motivation and highlight the advantages of technology for learning.

The fourth also portrays a negative and insignificant relationship between the variables, i.e. Perceived challenges and AT ($\beta = -0.054$; p -value > 0.1). Therefore, an increase in perceived challenges will abate students' attitudes toward adopting technology for MCPs. The findings were consistent with Almahasees and Qassem (2021). The findings conclude that students who encounter countless challenges with these programs are less enthusiastic about using technology for micro-credentials. Several factors could hinder students' attitudes toward using technology; for instance, students may confront technological challenges while using technology for MCPs, such as software problems, connectivity issues, or unfamiliarity with digital technologies. Secondly, students may develop negative attitudes, when students are unable to connect the use of technology and academic achievement or career goals. Therefore, these challenges should be addressed through proper support, instructional material and technological interfaces that can help improve students' attitudes toward technology utilization for MCPs.

The findings validate that students face various challenges with online learning programs, e.g. technological concerns, obtaining proper Internet connectivity, managing

time, organizing assignments and tasks, lack of engagement and interaction and identification issues. These challenges trigger students' reluctance to adopt technology for MCPs.

The Fifth hypothesis shows that PEOU significantly influences students' attitudes toward using technology for MCPs ($\beta = 0.133^{**}$; p -value < 0.05). Findings were consistent with studies of [Glavee-Geo et al. \(2017\)](#) and [Fianto et al. \(2020\)](#), results established PEOU as one essential element students evaluate while using technology for studying and learning. This implies that user-friendliness increases students' intentions toward using technology for such MCPs. The use of technology for MCPs positively influences students' behavioral intentions; thus, indicating awareness will enhance individuals' intentions toward using MCPs. According to [Asmy et al. \(2019\)](#), the ease of using technology strongly affects individuals' learning intentions in Pakistan, as it plays an important role in shaping their perspectives.

The sixth hypothesis indicates a positive and significant relationship between PL and attitude toward using technology ($\beta = 0.234^{***}$; p -value < 0.01). Findings are consistent with studies by [Lin \(2018\)](#) and [Babar \(2020\)](#) that determine several elements such as perceived interaction, course content, convenience, motivation, technical advancement and role of the teacher, are responsible for positive PL outcomes during using technology for MCPs. Thus, higher PL outcomes for using technology for MCPs will significantly increase students' behavioral intentions. Accordingly, educational institutions and instructors should concentrate on achieving learning outcomes through using MCPs.

The seventh hypothesis indicates a positive and significant relationship between PUn and AT ($\beta = 0.278^{***}$; p -value < 0.01). Findings were consistent with the studies of [Acheampong and Siiba \(2020\)](#), the findings suggest that students are highly interested in technology and believe it is a great tool for learning and improving their skills. In addition, students also ascertain that technology integration for learning will improve their productivity and portfolio. Furthermore, students' intention toward using technology for learning is directly connected with their attitude and intention of its usefulness. Thus, Students perceive technology integration for online programs as easy, exciting, time-saving, comfortable and fun. As a result, students will demonstrate behavioral intention toward using technology for MCPs for learning.

The LMCs do not moderate the relationship between attitude toward using technology and students' behavioral intentions. Thus, [H7](#) was not supported ($\beta = 0.020$; p -value < 0.1). The results are consistent with studies of [Agudo-Peregrina et al. \(2014\)](#). As students and institutions are unaware of MCPs; therefore they do not use technology for these programs. Furthermore, It implies that LMC have a negligible impact on students' perceptions of technology and their behavioral intentions ([Jackson and Tomlinson, 2020](#)). Despite the current condition of the labor market, students' attitudes toward using technology for academic purposes (such as MCPs) have a predictable and direct influence on their intentions to use technology for MCPs.

5. Conclusion and recommendations

5.1 Conclusion

Globally, MCPs have gained popularity as it provides credentials and learning activities which further lead to credentials, badges and rewards ([Cirlan and Loukkola, 2020](#)). Micro-credentials, particularly digital badges, are relatively new ideas in education and learning. While there has been substantial research on acceptability and use of technology in education, particular features (such as self-guide, digital badges, flexibility and certifications) and motives (such as highly accessible, economical learning opportunities and e-learning) associated with micro-credentials have not been properly investigated.

Students are increasingly reliant and enthusiastic about technology integration when they acknowledge various benefits it offers; for instance, increased creativity, professional growth, information acquisition, skill and expertise enhancement and lifetime learning. However, in Pakistan, students face challenges to use technology for micro-credentials owing to a lack of awareness and expertise. Therefore, this research study aims to investigate students' attitudes toward using technology for MCPs for learning. In addition, the study has incorporated LMC as a moderator to acquire an in-depth understanding of students' attitudes and behavioral intentions to use technology for MCPs. Furthermore, PLS-SEM is used with the Smart-PLS 3 software to evaluate the obtained data. The findings revealed that all the hypotheses were supported except for hypotheses H3 and H4. Students who are forced or pressured to use technology for these programs may develop a more negative perspective of technology in their learning experiences. Similarly, students who encounter greater challenges with these programs will be less enthusiastic about using technology for micro-credentials.

Furthermore, the findings indicate that LMC have a negligible effect on the relationship between attitude toward adopting technology for MCPs and behavioral intention. Therefore, benefactors of digital credential programs and higher education institutions should encourage, regulate and promote such MCPs. Furthermore, students should be encouraged to accept and participate in MCPs by providing them with sufficient awareness regarding the credentials, badges and awards they will receive. Because the world is moving at a fast pace and technology is getting more advanced day by day if students and countries like Pakistan do not accept the change will be left behind in the future. As a result, improving the levels of PEOU will increase the intention of students to adopt MCPs.

5.2 Theoretical contributions

The study contributes to research in the following ways. Firstly, the study aims to investigate students' attitudes toward using technology for MCPs and their behavioral intention toward using MCPs for learning and enhancing their skills. Secondly, this study has targeted the students of private sector universities because private sector universities focus on MCPs, so it was easy to collect data from these universities. Also, private-sector universities usually have stronger ties to companies and employers. These universities may strengthen their contacts with relevant firms and increase students' networking opportunities by giving micro-credentials targeted to specific industry demands (Miller and Jorre De St Jorre, 2022). Thirdly, two theoretical models were utilized to examine student attitudes (i.e. TAM and Self-determination). These models provide significant theoretical support for the proposed conceptual model that specified technology characteristics. Moreover, according to the author's knowledge, less number of studies have focused on the integration of these models together in examining the student's intention to use technology for MCPs. However, the PUn of technology is the most important factor influencing students' attitudes towards using technology for MCPs and it further encourages behavioral intention. Furthermore, the study provides evidence that PUn, PEOU and PL influenced students' attitudes to use technology in MCPs. The TAM framework is well-developed and validated in research and our findings corroborate the original TAM's significance for all variables. As a result, this TAM extension form is a substantial foundation for future MCP adoption studies; irrespective of certain extra external elements.

5.3 Managerial implications

The current study also provides some guidelines for students, universities, other educational institutes and digital credential program providers. This study will motivate students and individuals to gain additional skills and certifications through MCPs; it will

also help in understanding the factors that influence their desire and motivation to use technology. Additionally, the technology acceptance and self-determination model can assist people to make informed decisions about participation in technology-based MCPs to maximize the benefits and outcomes (Alsobhi *et al.*, 2023). The study finding will provide guidelines for students to learn new skills, improve their knowledge and develop abilities that apply to their present or desired employment positions. These programs can also assist students in career advancement, increased job satisfaction and personal and professional achievement and progress. The study findings conclude that students are more motivated to use technology-based education programs and MCPs when they consider micro-credentials as beneficial, flexible, easily comprehensible and high-quality learning programs.

The adoption of technology can substantially benefit universities, colleges and other educational institutions that offer MCPs. Initially, technology-based MCPs allowed universities to deliver courses efficiently with a larger number of students. Technology integration also allows collaboration with industry partners to develop relevant and up-to-date micro-credentials by using their experience and resources. Furthermore, universities should provide a platform for students to connect and create awareness regarding the system of MCPs. Also, universities should critically evaluate the challenges students face while using technology for MCPs; understanding and addressing these challenges through proper support, instructional material and technological interfaces can significantly improve students' attitudes toward using technology for MCPs. As proactive users of cutting-edge technology, educational institutions will gain a competitive edge as a result of this (Al-Kurdi *et al.*, 2020). Furthermore, it enhances a technologically enhanced learning environment, minimizing the overall cost of education.

Alternatively, other educational institutions could consider the results of this study to determine the needs and issues that students have with using technology for MCPs. Because micro-credentials, particularly digital badges, are still relatively new concepts in education and learning, educational institutions must focus on this to ensure that the implementation and accreditation of these credentials are properly addressed. Moreover, stakeholders of digital credential programs can also benefit from this adoption owing to increasing demand for powerful and user-friendly platforms and the incorporation of technology in corporations and educational institutions. This allows digital credential program providers to offer innovative solutions, make income and grow their market position.

The implications for research also provide valuable insights for other countries except Pakistan. Firstly, researchers can investigate how cultural differences influence the association among attitudes to use technology for micro-credentials and behavioral intention by conducting the study in multiple nations or regions. This cross-cultural technique helps in determining if observed patterns are constant across multiple populations or if there are variations due to cultural influences. Secondly, understanding the role of LMC as a moderator in different locations can assist policymakers in addressing specific workforce difficulties encountered by different countries, ensuring that workforce development measures correspond with the unique economic conditions of each region. Third, the study's findings can help technology providers and HEIs that offer MCPs by identifying how cultural factors influence attitudes and intentions may assist guide the design and marketing of technology-based learning platforms, allowing them to better suit the requirements and preferences of learners from diverse cultural backgrounds. Finally, the study's findings have the potential to have a significant impact on global employment. The findings can be used by policymakers and educators to create MCPs that fit with the skills in demand in different locations, making the workforce more appealing to employers and improving global economic competitiveness.

5.4 Limitations and future recommendations

The study has a few challenges for future researchers. Firstly, data collection was gathered from students in Karachi; thus, we suggested including students from other cities in the data collection process. Secondly, due to time constraints, this study is limited to acquiring responses cross-sectional, leading to limited inferences concerning the outputs. Therefore, future studies must adopt a longitudinal method to cross-validate the findings to overcome it. Furthermore, this study performs path analysis; therefore, future studies are recommended to check the mediation effect between the variables. Thirdly, this study is executed in a single city, i.e. Karachi, Pakistan; results should not be generalized without further testing. Therefore, it is recommended for future studies that they can conduct further testing by collecting samples from other cities and regions. Fourth, a limited set of variables were used due to specific research objectives; therefore, future studies may include additional variables. The TAM and the Self-determination model were utilized in this study. The researchers are recommended to investigate other models to observe different aspects that affect the acceptance of MCPs and the use of technologies for better education. Lastly, this study is a quantitative empirical study. Hence, we recommend using a qualitative or mixed-method research approach for future researchers.

References

- Abbasi, S., Ayoob, T., Malik, A. and Memon, S.I. (2020), "Perceptions of students regarding E-learning during Covid-19 at a private medial collage", *Pacific Journal of Medical Sciences*, Vol. 1 No. 1, pp. S57-S61.
- Abdullah, F. and Ward, R. (2016), "Developing a general extended technology acceptance model for E-learning (GETAMEL) by analysing commonly used external factors", *Computers in Human Behavior*, Vol. 56, pp. 238-256.
- Aboagye, E., Yawson, J.A. and Appiah, K.N. (2021), "COVID-19 and E-learning: the challenges of students in tertiary institutions", *Social Education Research*, Vol. 1 No. 1, pp. 1-8.
- Acheampong, R.A. and Siiba, A. (2020), "Modelling the determinants of car-sharing adoption intentions among young adults: the role of attitude, perceived benefits, travel expectations and socio-demographic factors", *Transportation*, Vol. 47 No. 5, pp. 2557-2580.
- Agudo-Peregrina, Á.F., Hernández-García, Á. and Pascual-Miguel, F.J. (2014), "Behavioral intention, use behavior and the acceptance of electronic learning systems: differences between higher education and lifelong learning", *Computers in Human Behavior*, Vol. 34, pp. 301-314.
- Aguilera-Hermida, A.P. (2020), "College students' use and acceptance of emergency online learning due to COVID-19", *International Journal of Educational Research Open*, Vol. 1, 100011.
- Al Kurdi, B., Alshurideh, M. and Salloum, S.A. (2020), "Investigating a theoretical framework for e-learning technology acceptance", *International Journal of Electrical and Computer Engineering (IJECE)*, Vol. 10 No. 6, pp. 6484-6496.
- Al Lily, A.E., Ismail, A.F., Abunasser, F.M. and Alqahtani, R.H.A. (2020), "Distance education as a response to pandemics: coronavirus and Arab culture", *Technology in Society*, Vol. 63, 101317.
- Al-Mamary, Y.H.S. (2022), "Understanding the use of learning management systems by undergraduate university students using the UTAUT model: credible evidence from Saudi Arabia", *International Journal of Information Management Data Insights*, Vol. 2 No. 2, 100092.
- Al-Maroofof, R.S., Salloum, S.A., Hassanien, A.E. and Shaalan, K. (2020), "Fear from COVID-19 and technology adoption: the impact of Google Meet during Coronavirus pandemic", *Interactive Learning Environments*, Vol. 31 No. 3, pp. 1293-1308.

- Almahasees, Z. and Qassem, M. (2021), "Faculty perception of teaching translation courses online during COVID-19", *PSU Research Review*, Vol. 6 No. 3, pp. 205-219.
- Alamer, A. and Al Khateeb, A. (2023), "Effects of using the WhatsApp application on language learners motivation: a controlled investigation using structural equation modelling", *Computer Assisted Language Learning*, Vol. 36 Nos 1-2, pp. 149-175.
- Alavi, M., Marakas, G.M. and Yoo, Y. (2002), "A comparative study of distributed learning environments on learning outcomes", *Information Systems Research*, Vol. 13 No. 4, pp. 404-415.
- Alea, L.A., Fabrea, M.F., Roldan, R.D.A. and Farooqi, A.Z. (2020), "Teachers' Covid-19 awareness, distance learning education experiences and perceptions towards institutional readiness and challenges", *International Journal of Learning, Teaching and Educational Research*, Vol. 19 No. 6, pp. 127-144.
- Ali, M. and Raza, S.A. (2017), "Service quality perception and customer satisfaction in Islamic banks of Pakistan: the modified SERVQUAL model", *Total Quality Management and Business Excellence*, Vol. 28 Nos 5-6, pp. 559-577.
- Ali, M., Raza, S.A. and Puah, C.-H. (2017), "Factors affecting to select Islamic credit cards in Pakistan: the TRA model", *Journal of Islamic Marketing*, Vol. 8 No. 3, pp. 330-344, doi: [10.1108/JIMA-06-2015-0043](https://doi.org/10.1108/JIMA-06-2015-0043).
- Alghizzawi, M., Habes, M., Salloum, S.A., Ghani, M.A., Mhamdi, C. and Shaalan, K. (2019), "The effect of social media usage on students' e-learning acceptance in higher education: a case study from the United Arab Emirates", *International Journal of Information Technology and Language Studies*, Vol. 3 No. 3, pp. 13-26.
- Ali, M., Puah, C.-H., Ali, A., Raza, S.A. and Ayob, N. (2022a), "Green intellectual capital, green HRM and green social identity toward sustainable environment: a new integrated framework for Islamic banks", *International Journal of Manpower*, Vol. 43 No. 3, pp. 614-638.
- Ali, M., Raza, S.A., Hakim, F., Puah, C.H. and Chaw, L.Y. (2022b), "An integrated framework for mobile payment in Pakistan: drivers, barriers, and facilitators of usage behavior", *Journal of Financial Services Marketing*, pp. 1-19.
- Almaiah, M.A., Al-Khasawneh, A. and Althunibat, A. (2020), "Exploring the critical challenges and factors influencing the E-learning system usage during COVID-19 pandemic", *Education and Information Technologies*, Vol. 25 No. 6, pp. 5261-5280.
- Al-Okaily, M., Lutfi, A., Alsaad, A., Taamneh, A. and Alsyouf, A. (2020), "The determinants of digital payment systems' acceptance under cultural orientation differences: the case of uncertainty avoidance", *Technology in Society*, Vol. 63, p. 101367.
- Al-Rahmi, W.M., Othman, M.S. and Musa, M.A. (2014), "The improvement of students' academic performance by using social media through collaborative learning in Malaysian higher education", *Asian Social Science*, Vol. 10 No. 8, p. 210.
- Alsobhi, H.A., Alakhtar, R.A., Ubaid, A., Hussain, O.K. and Hussain, F.K. (2023), "Blockchain-based micro-credentialing system in higher education institutions: systematic literature review", *Knowledge-Based Systems*, Vol. 265, 110238.
- Alvarez-Gonzalez, P., Lopez-Miguens, M.J. and Caballero, G. (2017), "Perceived employability in university students: developing an integrated model", *Career Development International*, Vol. 22 No. 3, pp. 280-299.
- Arellano-Bover, J. (2022), "The effect of labor market conditions at entry on workers' long-term skills", *Review of Economics and Statistics*, Vol. 104 No. 5, pp. 1028-1045.
- Arvanitis, A. and Kalliris, K. (2020), "Consistency and moral integrity: a self-determination theory perspective", *Journal of Moral Education*, Vol. 49 No. 3, pp. 316-329.
- Asmy, M., Mohd, B., Thaker, T., Bin, A., Pitchay, A., Bin, H. and Thas, M. (2019), "Factors influencing consumers' adoption of Islamic mobile banking services in Malaysia: an approach of partial least squares (PLS)", *Journal of Islamic Marketing*, Vol. 10 No. 4, pp. 1037-1056.

- Attard, C. and Holmes, K. (2022), "An exploration of teacher and student perceptions of blended learning in four secondary mathematics classrooms", *Mathematics Education Research Journal*, Vol. 34 No. 4, pp. 719-740.
- Babar, Y. (2020), *Essays on Online-Offline Interactions*, University of Minnesota Digital Conservancy, available at: <https://hdl.handle.net/11299/216418>
- Bailey, D.R., Almusharraf, N. and Almusharraf, A. (2022), "Video conferencing in the e-learning context: explaining learning outcome with the technology acceptance model", *Education and Information Technologies*, Vol. 27 No. 6, pp. 7679-7698.
- Bejaković, P. and Mrnjavac, Ž. (2020), "The importance of digital literacy on the labour market", *Employee Relations: The International Journal*, Vol. 42 No. 4, pp. 921-932.
- Belawati, T. (2014), "Open education, open education resources, and massive open online courses", *International Journal of Continuing Education and Lifelong Learning*, Vol. 7 No. 1, pp. 1-15.
- Bowers, J. and Kumar, P. (2015), "Students' perceptions of teaching and social presence: a comparative analysis of face-to-face and online learning environments", *International Journal of Web-Based Learning and Teaching Technologies (IJWLTT)*, Vol. 10 No. 1, pp. 27-44.
- Brown, M., Mhichil, M.N.G., Beirne, E. and Mac Lochlainn, C. (2021), "The global micro-credential landscape: charting a new credential ecology for lifelong learning", *Journal of Learning for Development*, Vol. 8 No. 2, pp. 228-254.
- Camilleri, M.A. (2021), "Evaluating service quality and performance of higher education institutions: a systematic review and a post-COVID-19 outlook", *International Journal of Quality and Service Sciences*, Vol. 13 No. 2, pp. 268-281.
- Carmines, E.G. and Zeller, R.A. (1979), *Reliability and Validity Assessment*, Sage publications.
- Chatterjee, S., Bhattacharjee, K.K., Tsai, C.W. and Agrawal, A.K. (2021), "Impact of peer influence and government support for successful adoption of technology for vocational education: a quantitative study using PLS-SEM technique", *Quality and Quantity*, Vol. 55 No. 6, pp. 2041-2064.
- Chen, Y., Dai, Y., Liu, A., Liu, W. and Jia, L. (2021), "Can the COVID-19 risk perception affect tourists' responsible behavior intention: an application of the structural equation model", *Journal of Sustainable Tourism*, Vol. 31 No. 9, pp. 2042-2061.
- Chen, H.R. and Tseng, H.F. (2012), "Factors that influence acceptance of web-based e-learning systems for the in-service education of junior high school teachers in Taiwan", *Evaluation and Program Planning*, Vol. 35 No. 3, pp. 398-406.
- Cheng, S.C. and Lai, C.L. (2020), "Facilitating learning for students with special needs: a review of technology-supported special education studies", *Journal of Computers in Education*, Vol. 7 No. 2, pp. 131-153.
- Chin, W.W. (1998), "The partial least squares approach to structural equation modeling", *Modern Methods for Business Research*, Vol. 295 No. 2, pp. 295-336.
- Chiu, W. and Cho, H. (2021), "The role of technology readiness in individuals' intention to use health and fitness applications: a comparison between users and non-users", *Asia Pacific Journal of Marketing and Logistics*, Vol. 33 No. 3, pp. 807-825.
- Cirlan, E. and Loukkola, T. (2020), "Micro-credentials linked to the Bologna key commitments", available at: <https://eua.eu/resources/publications/940/micro-credentials-linked-to-the-bologna-key-commitments.html>
- Clinefelter, D. and Aslanian, C. (2016), *Online College Students 2016: Comprehensive Data on Demands and Preferences*, The Learning House.
- Coman, C., Țiru, L.G., Meseșan-Schmitz, L., Stanciu, C. and Bularca, M.C. (2020), "Online teaching and learning in higher education during the coronavirus pandemic: students' perspective", *Sustainability*, Vol. 12 No. 24, 10367.

- Davis, F.D. (1989), "Perceived usefulness, perceived ease of use, and user acceptance of information technology", *MIS Quarterly*, Vol. 13 No. 3, pp. 319-340.
- Davis, F.D., Bagozzi, R.P. and Warshaw, P.R. (1989), "User acceptance of computer technology: a comparison of two theoretical models", *Management Science*, Vol. 35 No. 8, pp. 982-1003.
- De Naeghel, J., Van Keer, H., Vansteenkiste, M., Haerens, L. and Aelterman, N. (2016), "Promoting elementary school students' autonomous reading motivation: effects of a teacher professional development workshop", *The Journal of Educational Research*, Vol. 109 No. 3, pp. 232-252.
- Deci, E.L. and Ryan, R.M. (1985), "The general causality orientations scale: self-determination in personality", *Journal of Research in Personality*, Vol. 19 No. 2, pp. 109-134.
- Deci, E.L., Vallerand, R.J., Pelletier, L.G. and Ryan, R.M. (1991), "Motivation and education: the self-determination perspective", *Educational Psychologist*, Vol. 26 Nos 3-4, pp. 325-346.
- Dehghani, S., Sheikhi Fini, A.A., Zeinalipour, H. and Rezaei, E. (2020), "The competencies expected of instructors in massive open online courses (MOOCs)", *Interdisciplinary Journal of Virtual Learning in Medical Sciences*, Vol. 11 No. 2, pp. 69-83.
- Favale, T., Soro, F., Trevisan, M., Drago, I. and Mellia, M. (2020), "Campus traffic and e-Learning during COVID-19 pandemic", *Computer Networks*, Vol. 176, 107290.
- Fernandes, T. and Oliveira, E. (2021), "Understanding consumers' acceptance of automated technologies in service encounters: drivers of digital voice assistants adoption", *Journal of Business Research*, Vol. 122, pp. 180-191.
- Fianto, B.A., Hendratmi, A. and Aziz, P.F. (2020), "Factors determining behavioral intentions to use Islamic financial technology: three competing models", *Journal of Islamic Marketing*, Vol. 12 No. 4, pp. 794-812.
- Fishbein, M. and Ajzen, I. (1977), "Belief, attitude, intention, and behavior: an introduction to theory and research", *Philosophy and Rhetoric*, Vol. 10 No. 2, pp. 130-132.
- Fornell, C. and Larcker, D.F. (1981), "Structural equation models with unobservable variables and measurement error: algebra and statistics", *Journal of Marketing Research*, Vol. 18 No. 3, pp. 382-388, doi: [10.2307/3150980](https://doi.org/10.2307/3150980).
- Fu, X.M., Bao, Q., Xie, H. and Fu, X. (2021), "Diffusion of industrial robotics and inclusive growth: labour market evidence from cross country data", *Journal of Business Research*, Vol. 122, pp. 670-684.
- Gagne, M. and Deci, E.L. (2005), "Self-determination theory and work motivation", *Journal of Organizational Behavior*, Vol. 26 No. 4, pp. 331-362.
- Gamma, K., Mai, R., Cometta, C. and Loock, M. (2021), "Engaging customers in demand response programs: the role of reward and punishment in customer adoption in Switzerland", *Energy Research and Social Science*, Vol. 74, 101927.
- Gauthier, T. (2020), "The value of microcredentials: the employer's perspective", *The Journal of Competency-Based Education*, Vol. 5 No. 2, p. e01209.
- Gefen, D. and Straub, D. (2005), "A practical guide to factorial validity using PLS-Graph: tutorial and annotated example", *Communications of the Association for Information systems*, Vol. 16 No. 1, pp. 91-109.
- Ginting, D. (2021), "Student engagement and factors affecting active learning in English language teaching", *VELES (Voices of English Language Education Society)*, Vol. 5 No. 2, pp. 215-228.
- Glavee-Geo, R., Shaikh, A.A. and Karjaluo, H. (2017), "Mobile banking services adoption in Pakistan: are there gender differences?", *International Journal of Bank Marketing*, Vol. 35 No. 7, pp. 1090-1114.
- Gold, A.H., Malhotra, A. and Segars, A.H. (2001), "Knowledge management: an organizational capabilities perspective", *Journal of Management Information Systems*, Vol. 18 No. 1, pp. 185-214.

- Grant, A.M., Nurmohamed, S., Ashford, S.J. and Dekas, K. (2011), "The performance implications of ambivalent initiative: the interplay of autonomous and controlled motivations", *Organizational Behavior and Human Decision Processes*, Vol. 116 No. 2, pp. 241-251.
- Hagger, C. and Chatzisarantis, N.L. (2016), "The trans-contextual model of autonomous motivation in education: conceptual and empirical issues and meta-analysis", *Review of Educational Research*, Vol. 86 No. 2, pp. 360-407.
- Haines, V.Y., Jalette, P. and Larose, K. (2010), "The influence of human resource management practices on employee voluntary turnover rates in the Canadian non-governmental sector", *Industrial and Labor Relations Review*, Vol. 63, pp. 228-246.
- Hair, J.F., Ringle, C.M. and Sarstedt, M. (2011), "PLS-SEM: Indeed a silver bullet", *Journal of Marketing Theory and Practice*, Vol. 19 No. 2, pp. 139-152.
- Hair, Sarstedt, M., Hopkins, L. and Kuppelwieser, V.G. (2014), "Partial least squares structural equation modeling (PLS-SEM): an emerging tool in business research", *European Business Review*, Vol. 26 No. 2, pp. 106-121.
- Haleem, A., Javaid, M., Qadri, M.A. and Suman, R. (2022), "Understanding the role of digital technologies in education: a review", *Sustainable Operations and Computers*, Vol. 3, pp. 275-285.
- Hartnett, M.K. (2015), "Influences that undermine learners' perceptions of autonomy, competence and relatedness in an online context", *Australasian Journal of Educational Technology*, Vol. 31 No. 1.
- Hartwig, T., Ikkatai, Y., Takanashi, N. and Yokoyama, H.M. (2022), *Artificial Intelligence ELSI Score for Science and Technology: A Comparison between Japan and the US*, AI & SOCIETY, Canada, pp. 1-18.
- Healy, M. (2021), "Microcredential learners need quality careers and employability support", *Journal of Teaching and Learning for Graduate Employability*, Vol. 12 No. 1, pp. 21-23.
- Huang, F., Teo, T. and Zhou, M. (2020), "Chinese students' intentions to use the Internet-based technology for learning", *Educational Technology Research and Development*, Vol. 68 No. 1, pp. 575-591.
- Hunt, H.K. (1977), "CS/D-overview and future research directions", *Conceptualization and Measurement of Consumer Satisfaction and Dissatisfaction*, Marketing Science Institute, Berlin: Cambridge, MA, pp. 455-488.
- Ifinedo, P. (2018), "Determinants of students' continuance intention to use blogs to learn: an empirical investigation", *Behaviour and Information Technology*, Vol. 37 No. 4, pp. 381-392.
- Jackson, D. and Tomlinson, M. (2020), "Investigating the relationship between career planning, proactivity and employability perceptions among higher education students in uncertain labour market conditions", *Higher Education*, Vol. 80 No. 3, pp. 435-455.
- Kennedy, D. (2006), *Writing and Using Learning Outcomes: A Practical Guide*, University College Cork, Ireland.
- Kenton, W. (2023), *Labor Market Explained: Theories and Who Is Included*, available at: <https://www.investopedia.com/terms/l/labor-market.asp>
- Kiiskilä, P., Hanafy, A. and Pirkkalainen, H. (2022), "Features of micro-credential platforms in higher education", *14th International Conference on Computer Supported Education*, Vol. 1, pp. 81-91.
- Kiiskilä, P., Kukkonen, A. and Pirkkalainen, H. (2023), "Are micro-credentials valuable for students? Perspective on verifiable digital credentials", *SN Computer Science*, Vol. 4 No. 4, p. 366.
- Lai, H.M., Hsieh, P.J., Uden, L. and Yang, C.H. (2021), "A multilevel investigation of factors influencing university students' behavioral engagement in flipped classrooms", *Computers and Education*, Vol. 175, 104318.
- Lan, M. and Hew, K.F. (2020), "Examining learning engagement in MOOCs: a self-determination theoretical perspective using mixed method", *International Journal of Educational Technology in Higher Education*, Vol. 17 No. 1, pp. 1-24.

- Lee, M.C. (2009), "Factors influencing the adoption of internet banking: an integration of TAM and TPB with perceived risk and perceived benefit", *Electronic Commerce Research and Applications*, Vol. 8 No. 3, pp. 130-141.
- Lee, H.Y., Ahn, H. and Han, I. (2007), "VCR: virtual community recommender using the technology acceptance model and the user's needs type", *Expert Systems with Applications*, Vol. 33 No. 4, pp. 984-995.
- Lin, G.Y. (2018), "Anonymous versus identified peer assessment via a Facebook-based learning application: effects on quality of peer feedback, perceived learning, perceived fairness, and attitude toward the system", *Computers and Education*, Vol. 116, pp. 81-92.
- Lin, L.C., Hung, I.C., Kinshuk and Chen, N.-S. (2019), "The impact of student engagement on learning outcomes in a cyber-flipped course", *Educational Technology Research and Development*, Vol. 67, pp. 1573-1591.
- Littlejohn, A. and Hood, N. (2019), *Reconceptualising Learning in the Digital Age: the [Un] Democratising Potential of MOOCs*, SpringerBriefs in Education, Singapore.
- Looi, C.K., Chan, S.W. and Wu, L. (2021), "Crisis and opportunity: transforming teachers from curriculum deliverers to designers of learning", *Radical Solutions for Education in a Crisis Context: COVID-19 As an Opportunity for Global Learning*, pp. 131-145.
- Luqman, A., Masood, A. and Ali, A. (2018), "An SDT and TPB-based integrated approach to explore the role of autonomous and controlled motivations in SNS discontinuance intention", *Computers in Human Behavior*, Vol. 85, pp. 298-307.
- Lust, G., Collazo, N.A.J., Elen, J. and Clarebout, G. (2012), "Content management systems: enriched learning opportunities for all?", *Computers in Human Behavior*, Vol. 28 No. 3, pp. 795-808.
- Mailizar, M., Burg, D. and Maulina, S. (2021), "Examining university students' behavioural intention to use e-learning during the COVID-19 pandemic: an extended TAM model", *Education and Information Technologies*, Vol. 26 No. 6, pp. 7057-7077.
- Marshall, S.J. and Marshall, S.J. (2018), "Open education: a parable of change in higher education", *Shaping the University of the Future: Using Technology to Catalyse Change in University Learning and Teaching*, pp. 213-251.
- McCrae, R.R., Kurtz, J.E., Yamagata, S. and Terracciano, A. (2011), "Internal consistency, retest reliability, and their implications for personality scale validity", *Personality and Social Psychology Review*, Vol. 15 No. 1, pp. 28-50.
- McNamara, A.J. and Sepasgozar, S.M. (2020), "Developing a theoretical framework for intelligent contract acceptance", *Construction Innovation*, Vol. 20 No. 3, pp. 421-445.
- Meet, R.K. and Kala, D. (2021), "Trends and future prospects in MOOC researches: a systematic literature review 2013-2020", *Contemporary Educational Technology*, Vol. 13 No. 3.
- Middleton, J.A. and Toluk, Z. (1999), "First steps in the development of an adaptive theory of motivation", *Educational Psychologist*, Vol. 34 No. 2, pp. 99-112.
- Miller, K.K. and Jorre De St Jorre, T. (2022), "Digital micro-credentials in environmental science: an employer perspective on valued evidence of skills", *Teaching in Higher Education*, pp. 1-17.
- Milligan, S. and Kennedy, G. (2017), "To what degree? Alternative micro-credentialing in a digital age", in James, R., French, S. and Kelly, P. (Eds), *Visions for the Future of Australian Tertiary Education*, 1st ed., Centre for the Study of Higher Education, University of Melbourne, pp. 41-54.
- Mital, M., Pani, A.K., Damodaran, S. and Ramesh, R. (2015), "Cloud based management and control system for smart communities: a practical case study", *Computers in Industry*, Vol. 74, pp. 162-172.
- Morgan, H. (2022), "Understanding thematic analysis and the debates involving its use", *The Qualitative Report*, Vol. 27 No. 10, pp. 2079-2090.

- Musingafi, M.C., Mapuranga, B., Chiwanza, K. and Zebron, S. (2015), "Challenges for open and distance learning (ODL) students: experiences from students of the Zimbabwe Open University", *Journal of Education and Practice*, Vol. 6 No. 18, pp. 59-66.
- Nakayoshi, Y., Takase, M., Niitani, M., Imai, T., Okada, M., Yamamoto, K. and Takei, Y. (2021), "Exploring factors that motivate nursing students to engage in skills practice in a laboratory setting: a descriptive qualitative design", *International Journal of Nursing Sciences*, Vol. 8 No. 1, pp. 79-86.
- Nandi, A. and Mehendale, S. (2022), "E-learning in formal education under forced conditions using SDT and TAM", *Cardiometry*, No. 22, pp. 268-276.
- Navanitha, M., Savita, K.S., Arshad, N.I., Isawasan, P., Adams, D., Ahmat, N.H.C. and Shariman, T.P.N.B.T. (2022a), "The preliminary investigation on micro-credentials practices in Malaysia", *2022 International Conference on Digital Transformation and Intelligence (ICDI)*, IEEE, December, pp. 272-277.
- Navanitha, M., Savita, K.S., Arshad, N.I., Isawasan, P., Adams, D., Ahmat, N.H.C. and Shariman, T.P.N.B.T. (2022b), "Survey instrument validation for a learner's motivation to attain digital badge for micro-credential courses: cognitive interview and validity test", *Journal of Human University Natural Sciences*, Vol. 49 No. 10, pp. 77-87.
- Nikou, S.A. and Economides, A.A. (2017), "Mobile-based assessment: integrating acceptance and motivational factors into a combined model of self-determination theory and technology acceptance", *Computers in Human Behavior*, Vol. 68, pp. 83-95.
- Oliver, B. (2019), "Making micro-credentials work for learners, employers and providers", available at: <https://dteach.deakin.edu.au/microcredentials>
- Orman, R., Şimşek, E. and Çakır, M.A.K. (2023), "Micro-credentials and reflections on higher education", *Higher Education Evaluation and Development*, Vol. ahead-of-print No. ahead-of-print.
- Ouyang, X., Li, Q. and Du, K. (2020), "How does environmental regulation promote technological innovations in the industrial sector? Evidence from Chinese provincial panel data", *Energy Policy*, Vol. 139, 111310.
- Panigrahi, R., Srivastava, P.R. and Panigrahi, P.K. (2021), "Effectiveness of e-learning: the mediating role of student engagement on perceived learning effectiveness", *Information Technology and People*, Vol. 37 No. 7, pp. 1840-1862.
- Pappas, I.O., Giannakos, M.N. and Mikalef, P. (2017), "Investigating students' use and adoption of with-video assignments: lessons learnt for video-based open educational resources", *Journal of Computing in Higher Education*, Vol. 29, pp. 160-177.
- Patil, H. and Undale, S. (2023), "Willingness of university students to continue using e-Learning platforms after compelled adoption of technology: test of an extended UTAUT model", *Education and Information Technologies*, pp. 1-23.
- Peltier, J.W., Drago, W. and Schibrowsky, J.A. (2003), "Virtual communities and the assessment of online marketing education", *Journal of Marketing Education*, Vol. 25 No. 3, pp. 260-276.
- Perez-Sanagustin, M., Hernandez-Correa, J., Gelmi, C., Hilliger, I. and Rodriguez, M. (2016), "Does taking a MOOC as a complement for remedial courses affect my learning outcomes? A pilot study on calculus", *European conference on technology enhanced learning*, Springer International Publishing, Lyon, pp. 221-233.
- Petty, R.E., Gleichner, F. and Baker, S.M. (2020), "Multiple roles for affect in persuasion", *Emotion and Social Judgments*, Garland Science, pp. 181-200.
- Pickard, L., Shah, D. and De Simone, J.J. (2018), "Mapping microcredentials across MOOC platforms", *2018 Learning with MOOCS (LWMOOCS)*, pp. 17-21.
- Pollard, V. and Vincent, A. (2022), "Micro-credentials: a postdigital counternarrative", *Postdigital Science and Education*, Vol. 4 No. 3, pp. 843-859.

- Pozón-López, I., Higuera-Castillo, E., Muñoz-Leiva, F. and Liébana-Cabanillas, F.J. (2021), "Perceived user satisfaction and intention to use massive open online courses (MOOCs)", *Journal of Computing in Higher Education*, Vol. 33 No. 1, pp. 85-120.
- Prensky, M. (2009), "H. sapiens digital: from digital immigrants and digital natives to digital wisdom", *Innovate: Journal of Online Education*, Vol. 5 No. 3.
- Qazi, W., Qazi, Z., Raza, S.A., Hakim Shah, F. and Khan, K.A. (2023), "Students' employability confidence in COVID-19 pandemic: role of career anxiety and perceived distress", *Journal of Applied Research in Higher Education*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1108/JARHE-02-2022-0072](https://doi.org/10.1108/JARHE-02-2022-0072).
- Qenani, E., MacDougall, N. and Sexton, C. (2014), "An empirical study of self-perceived employability: improving the prospects for student employment success in an uncertain environment", *Active Learning in Higher Education*, Vol. 15 No. 3, pp. 199-213.
- Queiroz, M.M. and Wamba, S.F. (2019), "Blockchain adoption challenges in supply chain: an empirical investigation of the main drivers in India and the USA", *International Journal of Information Management*, Vol. 46, pp. 70-82.
- Qureshi, M.A., Khaskheli, A., Qureshi, J.A., Raza, S.A. and Khan, K.A. (2023), "Factors influencing green purchase behavior among millennials: the moderating role of religious values", *Journal of Islamic Marketing*, Vol. 14 No. 6, pp. 1417-1437.
- Racero, F.J., Bueno, S. and Gallego, M.D. (2020), "Predicting students' behavioral intention to use open source software: a combined view of the technology acceptance model and self-determination theory", *Applied Sciences*, Vol. 10 No. 8, p. 2711.
- Rad, D., Egerau, A., Roman, A., Dughi, T., Balas, E., Maier, Ignat, S. and Rad, G. (2022), *A preliminary investigation of the technology acceptance model (TAM) in early childhood education and care, BRAIN. Broad Research in Artificial Intelligence and Neuroscience*, Vol. 13 No. 1, pp. 518-533.
- Rapanta, C., Botturi, L., Goodyear, P., Guàrdia, L. and Koole, M. (2020), "Online university teaching during and after the Covid-19 crisis: refocusing teacher presence and learning activity", *Postdigital Science and Education*, Vol. 2 No. 3, pp. 923-945.
- Raza, S.A. and Khan, K.A. (2022), "Knowledge and innovative factors: how cloud computing improves students' academic performance", *Interactive Technology and Smart Education*, Vol. 19 No. 2, pp. 161-183.
- Raza, S.A., Shah, N. and Ali, M. (2019), "Acceptance of mobile banking in Islamic banks: evidence from modified UTAUT model", *Journal of Islamic Marketing*, Vol. 10 No. 1, pp. 357-376, doi: [10.1108/JIMA-04-2017-0038](https://doi.org/10.1108/JIMA-04-2017-0038).
- Raza, S.A., Qazi, W. and Yousufi, S.Q. (2021), "The influence of psychological, motivational, and behavioral factors on university students' achievements: the mediating effect of academic adjustment", *Journal of Applied Research in Higher Education*, Vol. 13 No. 3, pp. 849-870.
- Reed, A. (2023), "Micro-credentials and the role of evidence: increasing the potential for learner-centeredness, inclusivity and an expansive model of assessment and credentialing", *The International Journal of Information and Learning Technology*, Vol. ahead-of-print No. ahead-of-print.
- Reinecke, L., Vorderer, P. and Knop, K. (2014), "Entertainment 2.0? The role of intrinsic and extrinsic need satisfaction for the enjoyment of Facebook use", *Journal of Communication*, Vol. 64 No. 3, pp. 417-438.
- Ringle, C., Da Silva, D. and Bido, D. (2015), "Structural equation modeling with the SmartPLS", *Brazilian Journal of Marketing*, Vol. 13 No. 2, pp. 56-73.
- Roca, J.C. and Gagné, M. (2008), "Understanding e-learning continuance intention in the workplace: a self-determination theory perspective", *Computers in Human Behavior*, Vol. 24 No. 4, pp. 1585-1604.

- Rodrigues, F., Jacinto, M., Couto, N., Monteiro, D., Monteiro, A.M., Forte, P. and Antunes, R. (2023), "Motivational correlates, satisfaction with life, and physical activity in older adults: a structural equation analysis", *Medicina*, Vol. 59 No. 3, p. 599.
- Ryan, R.M. and Connell, J.P. (1989), "Perceived locus of causality and internalization: examining reasons for acting in two domains", *Journal of Personality and Social Psychology*, Vol. 57 No. 5, p. 749.
- Ryan, R.M. and Deci, E.L. (2000), "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being", *American Psychologist*, Vol. 55 No. 1, p. 68.
- Ryan, R.M. and Deci, E.L. (2020), "Intrinsic and extrinsic motivation from a self-determination theory perspective: definitions, theory, practices, and future directions", *Contemporary Educational Psychology*, Vol. 61, 101860.
- Ryan, A.M., Hicks, L. and Midgley, C. (1997), "Social goals, academic goals, and avoiding seeking help in the classroom", *The Journal of Early Adolescence*, Vol. 17 No. 2, pp. 152-171.
- Sabah, N.M. (2020), "Motivation factors and barriers to the continuous use of blended learning approach using Moodle: students' perceptions and individual differences", *Behaviour and Information Technology*, Vol. 39 No. 8, pp. 875-898.
- Sadaf, A., Newby, T.J. and Ertmer, P.A. (2012), "Exploring factors that predict preservice teachers' intentions to use Web 2.0 technologies using decomposed theory of planned behavior", *Journal of Research on Technology in Education*, Vol. 45 No. 2, pp. 171-196.
- Sahu, A.K., Padhy, R.K. and Dhir, A. (2020), "Envisioning the future of behavioral decision-making: a systematic literature review of behavioral reasoning theory", *Australasian Marketing Journal (AMJ)*, Vol. 28 No. 4, pp. 145-159.
- Scherer, R., Siddiq, F. and Tondeur, J. (2019), "The technology acceptance model (TAM): a meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education", *Computers and Education*, Vol. 128, pp. 13-35.
- Schwab, K. (2016), "The fourth industrial revolution: what it means, how to respond", available at: www.weforum.org/agenda/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond/
- Sebastianelli, R., Swift, C. and Tamimi, N. (2015), "Factors affecting perceived learning, satisfaction, and quality in the online MBA: a structural equation modeling approach", *Journal of Education for Business*, Vol. 90 No. 6, pp. 296-305.
- Shah, D., Brady, B. and Pickard, L. (2019), *850+ Online Courses with Real College Credit that You Can Access for Free*, available at: <https://www.classcentral.com/report/free-for-credit-moocs/>
- Shao, Z. (2022), *The Adoption of Digital Badges in Higher Education: A Case Study of a Faculty Professional Development Program*, Indiana University, Indiana.
- Sharma, S.K., Gaur, A., Saddikuti, V. and Rastogi, A. (2017), "Structural equation model (SEM)-neural network (NN) model for predicting quality determinants of e-learning management systems", *Behaviour and Information Technology*, Vol. 36 No. 10, pp. 1053-1066.
- Sima, V., Gheorghe, I.G., Subić, J. and Nancu, D. (2020), "Influences of the industry 4.0 revolution on the human capital development and consumer behavior: a systematic review", *Sustainability*, Vol. 12 No. 10, p. 4035.
- Songkram, N., Chootongchai, S., Osuwan, H., Chuppunnarat, Y. and Songkram, N. (2023), "Students' adoption towards behavioral intention of digital learning platform", *Education and Information Technologies*, pp. 1-23.
- Sulaiman, A., Jaafar, N.I. and Aziz, N.A.A. (2012), "Factors influencing intention to use MYEPF I-Akaun", *World Applied Sciences Journal*, Vol. 18 No. 3, pp. 451-461.
- Tamilmani, K., Rana, N.P. and Dwivedi, Y.K. (2021), "Consumer acceptance and use of information technology: a meta-analytic evaluation of UTAUT2", *Information Systems Frontiers*, Vol. 23 No. 4, pp. 987-1005.

- Taylor, S. and Todd, P.A. (1995), "Understanding information technology usage: a test of competing models", *Information Systems Research*, Vol. 6 No. 2, pp. 144-176.
- Teo, T., Huang, F. and Hoi, C.K.W. (2018), "Explicating the influences that explain intention to use technology among English teachers in China", *Interactive Learning Environments*, Vol. 26 No. 4, pp. 460-475.
- Thi Ngoc Ha, N., Spittle, M., Watt, A. and Van Dyke, N. (2022), "A systematic literature review of micro-credentials in higher education: a non-zero-sum game", *Higher Education Research and Development*, Vol. 46 No. 2, pp. 1527-1548.
- Thong, J.Y., Hong, S.J. and Tam, K.Y. (2006), "The effects of post-adoption beliefs on the expectation-confirmation model for information technology continuance", *International Journal of Human-Computer Studies*, Vol. 64 No. 9, pp. 799-810.
- Van den Broeck, A., Howard, J.L., Van Vaerenbergh, Y., Leroy, H. and Gagné, M. (2021), "Beyond intrinsic and extrinsic motivation: a meta-analysis on self-determination theory's multidimensional conceptualization of work motivation", *Organizational Psychology Review*, Vol. 11 No. 3, pp. 240-273.
- Wayre, E. and Vonthron, A.-M. (2019), "Relational and psychological factors affecting exam participation and student achievement in online college courses", *The Internet and Higher Education*, Vol. 43, 100671.
- Venkatesh, V. and Bala, H. (2008), "Technology acceptance model 3 and a research agenda on interventions", *Decision Sciences*, Vol. 39 No. 2, pp. 273-315.
- Venkatesh, V. and Davis, F.D. (2000), "A theoretical extension of the technology acceptance model: four longitudinal field studies", *Management Science*, Vol. 46 No. 2, pp. 186-204.
- Venkatesh, V., Morris, M.G., Davis, G.B. and Davis, F.D. (2003), "User acceptance of information technology: toward a unified view", *MIS Quarterly*, pp. 425-478.
- Venkatesh, V., Thong, J.Y. and Xu, X. (2012), "Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology", *MIS Quarterly*, Vol. 36 No. 1, pp. 157-178.
- Virani, S.R., Saini, J.R. and Sharma, S. (2023), "Adoption of massive open online courses (MOOCs) for blended learning: the Indian educators' perspective", *Interactive Learning Environments*, Vol. 31 No. 2, pp. 1060-1076.
- Voßemer, J., Gebel, M., Täht, K., Unt, M., Högberg, B. and Strandh, M. (2018), "The effects of unemployment and insecure jobs on well-being and health: the moderating role of labor market policies", *Social Indicators Research*, Vol. 138, pp. 1229-1257.
- Wang, W.T. and Hou, Y.P. (2015), "Motivations of employees' knowledge sharing behaviors: a self-determination perspective", *Information and Organization*, Vol. 25 No. 1, pp. 1-26.
- Wang, M., Wang, M., Cui, Y. and Zhang, H. (2021), "Art teachers' attitudes toward online learning: an empirical study using self determination theory", *Frontiers in Psychology*, Vol. 12, 627095.
- Wheelahan, L. and Moodie, G. (2021), "Analysing micro-credentials in higher education: a Bernsteinian analysis", *Journal of Curriculum Studies*, Vol. 53 No. 2, pp. 212-228.
- Wong, K.T., Teo, T. and Russo, S. (2012), "Influence of gender and computer teaching efficacy on computer acceptance among Malaysian student teachers: an extended technology acceptance model", *Australasian Journal of Educational Technology*, Vol. 28 No. 7, pp. 1190-1207.
- Wu, Y. and Li, C. (2019), "Helping others helps? A self-determination theory approach on work climate and wellbeing among volunteers", *Applied Research in Quality of Life*, Vol. 14, pp. 1099-1111.
- Xu, L., Zhang, J., Ding, Y., Sun, G., Zhang, W., Philbin, S.P. and Guo, B.H. (2022), "Assessing the impact of digital education and the role of the big data analytics course to enhance the skills and employability of engineering students", *Frontiers in Psychology*, Vol. 13, 974574.
- Yu, S.Y. (2021a), "A review of the accessibility of ACT COVID-19 information portals", *Technology in Society*, Vol. 64, 101467.

- Yu, Z. (2021b), "The effects of gender, educational level, and personality on online learning outcomes during the COVID-19 pandemic", *International Journal of Educational Technology in Higher Education*, Vol. 18 No. 1, p. 14.
- Zhang, K.Z. and Benyoucef, M. (2016), "Consumer behavior in social commerce: a literature review", *Decision Support Systems*, Vol. 86, pp. 95-108.
- Zhou, M. (2016), "Chinese university students' acceptance of MOOCs: a self-determination perspective", *Computers and Education*, Vol. 92, pp. 194-203.

Further reading

- Ahmat, N.H.C., Bashir, M.A.A., Razali, A.R. and Kasolang, S. (2021), "Micro-credentials in higher education institutions: challenges and opportunities", *Asian Journal of University Education*, Vol. 17 No. 3, pp. 281-290.
- Baber, H. (2020), "Determinants of students' perceived learning outcome and satisfaction in online learning during the pandemic of COVID-19", *Journal of Education and E-Learning Research*, Vol. 7 No. 3, pp. 285-292.
- Cheon, J., Lee, S., Crooks, S.M. and Song, J. (2012), "An investigation of mobile learning readiness in higher education based on the theory of planned behavior", *Computers and Education*, Vol. 59 No. 3, pp. 1054-1064.
- Hu, P.J.H. and Hui, W. (2012), "Examining the role of learning engagement in technology-mediated learning and its effects on learning effectiveness and satisfaction", *Decision Support Systems*, Vol. 53 No. 4, pp. 782-792.
- Kahn, L.B. (2010), "The long-term labor market consequences of graduating from college in a bad economy", *Labour Economics*, Vol. 17 No. 2, pp. 303-316.
- Kuh, G.D. (2003), "What we're learning about student engagement from NSSE: benchmarks for effective educational practices", *Change: The Magazine of Higher Learning*, Vol. 35 No. 2, pp. 24-32.
- Oreopoulos, P., von Wachter, T. and Heisz, A. (2012), "The short- and long-term career effects of graduating in a recession", *American Economic Journal: Applied Economics*, Vol. 4 No. 1, pp. 1-29.
- Philpott, A. and Son, J.B. (2022), "Leaderboards in an EFL course: student performance and motivation", *Computers and Education*, Vol. 190, 104605.
- Prabhu, M., Bolar, N.B., Mallya, K., Roy, J., Payini, V.P. and Thirugnanasambantham, K. (2021), "Determinants of hospitality students' perceived learning during COVID 19 pandemic: role of interactions and self-efficacy", *Journal of Hospitality, Leisure.Sport and Tourism Education*, Vol. 30, 100335.
- Ryan, R.M. (2017), *Self-Determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness*, The Guilford Press A Division of Guilford Publications, New York.
- Saeed Al-Marouf, R., Alhumaid, K. and Salloum, S. (2020), "The continuous intention to use e-learning, from two different perspectives", *Education Sciences*, Vol. 11 No. 1, p. 6.
- Sailer, M., Stadler, M., Schultz-Pernice, F., Franke, U., Schöffmann, C., Paniotova, V., Husagic, L. and Fischer, F. (2021), "Technology-related teaching skills and attitudes: validation of a scenario-based self-assessment instrument for teachers", *Computers in Human Behavior*, Vol. 115, 106625.
- Serhan, D. (2020), "Transitioning from face-to-face to remote learning: students' attitudes and perceptions of using zoom during COVID-19 pandemic", *International Journal of Technology in Education and Science*, Vol. 4 No. 4, pp. 335-342.
- Shaw, J.D., Delery, J.E., Jenkins, G.D. and Gupta, N. (1998), "An organization-level analysis of voluntary and involuntary turnover", *Academy of Management Journal*, Vol. 41, pp. 511-525.
- Siahaan, A.U., Aji, S.B., Antoni, C. and Handayani, Y. (2020), "March. Online social learning platform vs E-learning for higher vocational education in purpose to English test preparation", *7th International Conference on English Language and Teaching (ICOELT 2019)*, Atlantis Press, pp. 76-82.

-
- Wu, B. and Zhang, C. (2014), "Empirical study on continuance intentions towards E-Learning 2.0 systems", *Behaviour and Information Technology*, Vol. 33 No. 10, pp. 1027-1038.
- Zaric, N., Lukarov, V. and Schroder, U. (2020), "A fundamental study for gamification design: exploring learning tendencies' effects", *International Journal of Serious Games*, Vol. 7 No. 4, pp. 3-25.

Corresponding author

Mansoor Ahmed can be contacted at: mansoorahmed@hotmail.com

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgroupublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com